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PD Dr. med. Dr. rer. physiol. Peter Dietrich
Medizinische Klinik 1
Universitätsklinikum Erlangen
Ref. he_030_PD

Re: Doktorand (m/w/d) — Bioinformatik, Hepatologie / Onkologie (Ref. he_030_PD)

Dear PD Dr. Dietrich,

I am applying for the doctoral position (Ref. he_030_PD) in bioinformatic analysis of human and murine tissues. The combination of single-cell and bulk RNA sequencing analysis with a hepatology and oncology focus describes precisely the analytical domain for which I have developed and validated computational tools over the past four years.

RNA-seq analysis infrastructure. The Gospel framework handles the full RNA-seq analysis pipeline: read alignment and quantification, differential expression analysis across conditions and cohorts, splicing analysis, and integration with variant calling outputs for somatic mutation-expression correlation. Validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank, it processes 10^6 variants per second with ClinVar, KEGG, and Reactome pathway annotation built in. For the comparison of human and murine transcriptomes — a methodological challenge specific to this project — Gospel handles species-aware analysis with ortholog mapping, differential gene expression across species, and cross-species pathway conservation testing. These comparisons are central to translational validation studies: establishing which molecular phenotypes identified in murine models are conserved in human patient tissue.

Single-cell sequencing analysis. For single-cell RNA-seq, my primary contribution is the statistical inference layer: variational Bayes posterior approximation at $\mathcal{O}(10^6)$ latent variable scale, probabilistic graphical models for multi-omics integration, and Gaussian mixture models for cell population structure. These methods sit directly below the standard Scanpy/Seurat pipeline — Leiden clustering and UMAP are dimensionality reduction steps whose statistical behaviour I understand from the underlying geometry. For hepatology-specific single-cell analysis this matters concretely: the liver contains many rare cell types (hepatic stellate cells, Kupffer cells, sinusoidal endothelial cells, periportal vs. pericentral hepatocytes) whose identification requires probabilistic models that handle low cell counts robustly, not just nearest-neighbour clustering. RNA velocity and diffusion pseudotime — trajectory inference for fibrogenesis progression, tumour microenvironment remodelling, or immune cell recruitment sequences — are directly supported by the Gaussian process infrastructure I have built.

Hepatocellular carcinoma immunobiology. The oncology component of this project most likely involves the hepatic tumour microenvironment: immune evasion, cytokine signalling between tumour and stromal cells, or HCC neoantigen biology. The Syndrome framework I have developed computes spectral complementarity between molecular sequences to predict MHC-I/II binding (AUC 0.81 vs. NetMHCpan 0.68 on 2,847 IEDB peptides), identify neoantigens from somatic mutation data, and score immune escape variants via single amino acid substitution scanning. For an RNA-seq study of HCC, this provides a direct bridge between gene expression data and immunopeptidological function: identifying which tumour-expressed transcripts generate immunogenic peptides, which mutant peptides escape T cell recognition, and how the immune state of the tumour microenvironment (quantified as the immune activation dimension D_{Imm} of the disease state vector $\mathbf{D} \in [0, 1]^8$) correlates with treatment response.

Molecular biology and in vivo experience. My computational background is primary, but it is grounded in wet lab experience. At Universitätsklinikum Hamburg-Eppendorf (2017–2018) I worked directly in the Glycomics and Proteomics laboratory: automated LC-MS/MS and MALDI-TOF pipeline operation, clinical sample processing, and data quality control for multi-centre clinical cohort studies. At TUM / Bayerisches Forschungsinstitut (2019–2021) I maintained computational pipelines processing MS omics data from clinical cohorts, working closely with the wet lab teams producing the data. I understand sample preparation constraints, how technical variation from different dissection or fixation protocols propagates into sequencing data quality, and how to flag analytical artefacts that originate in the biological protocol rather than the analysis. The murine tissue component of this project is an opportunity to extend that grounding to in vivo experimental design, which I pursue actively.

Background. I hold an MSc in Computational Biology (Universität Tübingen, 2018–2019) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg, 2014–2017), with doctoral research in Computational Lipidomics at TUM / Bayerisches Forschungsinstitut (2019–2021). I am legally resident in Germany with full work authorisation. English is native; German is conversational. I am available from 1 July 2026.

Yours sincerely,

Kundai Farai Sachikonye